

KARTHIK AGRAWAL

Computer Science Engineering Student | Backend, AI & Mobile Systems

CONTACT INFORMATION

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PROFESSIONAL SUMMARY

Computer Science Engineering student with hands-on experience in **Java backend development**, **REST - based web applications**, **JDBC**, **MySQL**, **Flutter mobile applications**, and **AI-based systems**. Built **end-to-end projects** involving **database design**, **authentication**, **computer vision**, and **API integration**. Strong foundation in **data structures**, **problem solving**, and **system design**. Actively seeking **internship opportunities** to apply and expand technical skills in real-world environments.

EDUCATION

- Bachelor of Engineering – Computer Science Engineering
Sri Venkateshwara College of Engineering, Bangalore
 - Class 12 – Sri Chaitanya Techno School
 - Class 10 – B.P. Indian Public School
- CGPA: 9.0 | 2023 – 2027
90% | 2022
85.5% | 2020

TECHNICAL SKILLS

Programming: Java, Dart (Flutter), Python, C

Backend: Java Servlets, JDBC, REST APIs, Spring Boot, Apache Tomcat

Frontend & Mobile: HTML, CSS, JavaScript, Flutter

Databases: MySQL, MongoDB, H2

AI / CV: OpenCV, TensorFlow, OCR, NLP, CNN

Tools: Git, GitHub, VS Code

Core CS Concepts:

Data Structures & Algorithms, DBMS, OOPs; Operating Systems, Computer Networks;
Software Engineering Fundamentals, SDLC, Basic System Design; Strong problem-solving skills.

PROJECTS

Primary Projects

1. **Online Examination System** :- {Java, Servlets, JDBC, MySQL}
 - Developed a **Java-based web application** with role-based authentication for administrators and students
 - Implemented **CRUD operations**, exam workflows, and **automated evaluation** using JDBC and MySQL
 - Deployed the application on **Apache Tomcat**
2. **DocuVerify System (AI + Blockchain)** {Python, OpenCV, TensorFlow, MongoDB}
 - Implemented **OCR pipelines** and NLP-based semantic validation for document verification

- Built **CNN-based models** to detect forged signatures, seals, and layout tampering
- Ensured **data integrity** using blockchain-based hash storage
- 3. **Sports Talent Assessment System (AI + Flutter)** {*Python, OpenCV, Flutter*}
 - Applied computer vision techniques to analyze athlete motion and posture from video input
 - Generated performance scores and detected invalid submissions
 - Built a **cross-platform Flutter frontend** for result visualization

Additional Projects

- **Farm-Tech System:** IoT-based monitoring of soil and water levels with a web dashboard and automated alerts
- **Screen Sense App:** Flutter-based application for real-time on-screen content analysis and AI-based verification

VIRTUAL EXPERIENCE

J.P. Morgan – Advanced Software Engineering Simulation (Forage)

- Completed a **J.P. Morgan–designed virtual software engineering program** on Forage, simulating industry-style development tasks
- Worked with an existing **codebase** by setting up the development environment, debugging broken code, and extending functionality
- Gained exposure to **data feeds, backend processing, and system-level thinking** in a fintech context

CERTIFICATIONS

- Java Developer Certification – Infosys SpringBoard
- Spring Boot with H2 Embedded Database – Coursera
- Prompt Engineering – Infosys SpringBoard
- AWS for Beginners – Great Learning Academy

ACHIEVEMENTS

- Maintained CGPA 9.0 in Computer Science Engineering
- Selected participant in Smart India Hackathon (SIH) for an AI-based Sports Talent Assessment System
- Successfully delivered multiple production-style academic and hackathon projects

ACTIVITIES & ENGAGEMENT

- Active member of Google Developer Student Clubs (GDSC)
- Participated in technical workshops on AI, backend development, and IoT
- Engaged in hackathons and coding competitions involving team-based problem solving
- Consistent self-driven learning through project-based exploration of new technologies